THE CATEGORY OF SCHEMES HAS FINITE LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3233-3335, following the statement at 3226-3231.

Editorial note. The proof opens with 'Please skip this proof', a pedagogical aside to readers, not a completeness disclaimer; the verification of (2)(a)-(2)(c) is complete.

Editorial note. Condition (2)(a) is concluded by the named affine fibre-product engine with the locally-ringed-spaces parenthetical reproduced.

Allowed standard prerequisites (editorial).

- Spec(Z) is final among locally ringed spaces. lemma-morphism-into-affine (same chapter, cited at ~3239-3241). Reduces existence of fibre products to representability of the fibre-product functor.
- Functor glueing framework. lemma-glue-functors (same chapter, card L26382 this batch, cited at ~3252-3253). The engine whose sheaf property and conditions (2)(a)-(2)(c) are verified for F.
- Mapping property of the glued locally ringed space. lemma-glue (same chapter, statement at 2761-2798, cited at ~3271). Justifies glueing the morphisms a_i into a: T -> X.
- Fibre products of affine schemes. lemma-fibre-product-affine-schemes (same chapter, statement at 1202, cited at ~3310). Engine for condition (2)(a); the parenthetical notes affine fibre products are fibre products in all locally ringed spaces.

Source reference. Original reference 002O: focused-seeds/geometry/S9-schemes-fibre-products/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01I4: sources/stacks-schemes/pinned-original-source.tex, source lines 1202–1202. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01JB: sources/stacks-schemes/pinned-original-source.tex, source lines 2762–2762. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:schemes.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01JJ: sources/stacks-schemes/pinned-original-source.tex, source lines 3110–3110. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01I1: sources/stacks-schemes/pinned-original-source.tex, source lines 1058–1058. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *The category of schemes has a final object, products and fibre products. In other words, the category of schemes has finite limits, see Categories, Lemma 002O.*

Proof. Please skip this proof. It is more important to learn how to work with the fibre product which is explained in the next section.

By Lemma 01I1 the scheme $\mathrm{Spec}(\mathbf{Z})$ is a final object in the category of locally ringed spaces. Thus it suffices to prove that fibred products exist.

Let $f : X \to S$ and $g : Y \to S$ be morphisms of schemes. We have to show that the functor

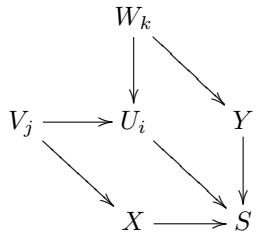
$$\begin{aligned} F : Sch^{opp} &\longrightarrow Sets \\ T &\longmapsto \mathrm{Mor}(T, X) \times_{\mathrm{Mor}(T,S)} \mathrm{Mor}(T, Y) \end{aligned}$$

is representable. We claim that Lemma 01JJ applies to the functor F. If we prove this then the lemma is proved.

First we show that F satisfies the sheaf property in the Zariski topology. Namely, suppose that T is a scheme, $T = \bigcup_{i \in I} U_i$ is an open covering, and $\xi_i \in F(U_i)$ such that $\xi_i|_{U_i \cap U_j} = \xi_j|_{U_i \cap U_j}$ for all pairs i, j. By definition ξ_i corresponds to a pair (a_i, b_i) where $a_i : U_i \to X$ and $b_i : U_i \to Y$ are morphisms of schemes such that $f \circ a_i = g \circ b_i$. The glueing condition says that $a_i|_{U_i \cap U_j} = a_j|_{U_i \cap U_j}$ and $b_i|_{U_i \cap U_j} = b_j|_{U_i \cap U_j}$. Thus by glueing the morphisms a_i we obtain a morphism of locally ringed spaces (i.e., a morphism of schemes) $a : T \to X$ and similarly $b : T \to Y$ (see for example the mapping property of Lemma 01JB). Moreover, on the members of an open covering the compositions $f \circ a$ and $g \circ b$ agree. Therefore $f \circ a = g \circ b$ and the pair (a, b) defines an element of $F(T)$ which restricts to the pairs (a_i, b_i) on each U_i. The sheaf condition is verified.

Next, we construct the family of subfunctors. Choose an open covering by open affines $S = \bigcup_{i \in I} U_i$. For every $i \in I$ choose open coverings by open affines $f^{-1}(U_i) = \bigcup_{j \in J_i} V_j$ and $g^{-1}(U_i) = \bigcup_{k \in K_i} W_k$. Note that $X = \bigcup_{i \in I} \bigcup_{j \in J_i} V_j$ is an open covering and similarly for Y. For any $i \in I$ and each pair $(j, k) \in J_i \times K_i$

we have a commutative diagram



where all the skew arrows are open immersions. For such a triple we get a functor

$$\begin{aligned} F_{i,j,k} : Sch^{opp} &\longrightarrow Sets \\ T &\longmapsto \mathrm{Mor}(T, V_j) \times_{\mathrm{Mor}(T,U_i)} \mathrm{Mor}(T, W_k). \end{aligned}$$

There is an obvious transformation of functors $F_{i,j,k} \to F$ (coming from the huge commutative diagram above) which is injective, so we may think of $F_{i,j,k}$ as a subfunctor of F.

We check condition (2)(a) of Lemma 01JJ. This follows directly from Lemma 01I4. (Note that we use here that the fibre products in the category of affine schemes are also fibre products in the whole category of locally ringed spaces.)

We check condition (2)(b) of Lemma 01JJ. Let T be a scheme and let $\xi \in F(T)$. In other words, $\xi = (a, b)$ where $a : T \to X$ and $b : T \to Y$ are morphisms of schemes such that $f \circ a = g \circ b$. Set $V_{i,j,k} = a^{-1}(V_j) \cap b^{-1}(W_k)$. For any further morphism $h : T' \to T$ we have $h^*\xi = (a \circ h, b \circ h)$. Hence we see that $h^*\xi \in F_{i,j,k}(T')$ if and only if $a(h(T')) \subset V_j$ and $b(h(T')) \subset W_k$. In other words, if and only if $h(T') \subset V_{i,j,k}$. This proves condition (2)(b).

We check condition (2)(c) of Lemma 01JJ. Let T be a scheme and let $\xi = (a, b) \in F(T)$ as above. Set $V_{i,j,k} = a^{-1}(V_j) \cap b^{-1}(W_k)$ as above. Condition (2)(c) just means that $T = \bigcup V_{i,j,k}$ which is evident. Thus the lemma is proved and fibre products exist. $\square$